

Forkfall

Player & Operator Guide — Testnet Alpha

Every step from an empty wallet to a settled ranked match: minting the soulbound starter deck, buying and opening packs, building and registering decks, matchmaking, playing a match, and settling it on-chain.

Version	MVP 0.1 · branch ccr-63eea11d-n5kv9d
Networks	Base Sepolia (84532, settlement hub) · Robinhood Chain testnet (46630) · local Anvil (31337)
Card set	Prototype set: 40 collectible cards (8 per race + 8 neutral)
Tools	Web app, Foundry (forge / cast), agent SDK, MCP server

Testnet only. Every contract reverts if deployed on a chain other than Anvil, Base Sepolia, Robinhood Chain testnet or Ethereum Sepolia, and every Foundry script and the server refuse other chains. Cards, packs and tokens here have no real value. Never use a wallet that holds real funds.

Contents

1. Getting set up
2. Deploying the contracts (operators)
3. Minting the soulbound starter deck
4. Buying and opening normal packs
5. Your collection and crafting
6. Deck builder
7. Identity: agents and verified humans
8. Matchmaking
9. Playing a match
10. The four races and keywords
11. After the match: settlement and ratings
12. Season rewards
13. Playing as an agent
14. Card reference
15. Troubleshooting

1. Getting set up

Forkfall has three ways in. Most players only need the first; on-chain actions (starter decks, packs, decks, settlement) go through Foundry.

SURFACE	USED FOR	WHERE
Web app	Practice, matchmaking, playing, watching, reading results	http://localhost:8787 (or your operator's URL)
Foundry scripts	Minting starters, buying/opening packs, registering decks and agents, settling matches, checking status	contracts/script/Play.s.sol
Agent SDK / MCP	Bots and AI agents playing through the same API as humans	packages/sdk, apps/mcp

Requirements

- Node 26 and pnpm 12.
- Foundry 1.8.3 (forge, cast, anvil).
- A **testnet-only** wallet private key with a little Base Sepolia ETH for gas (and for packs, if you pay in ETH).

Install

```
git clone --recursive https://github.com/gewubuntu/Forkfall.git && cd Forkfall
pnpm install
cp .env.example .env          # fill in TESTNET values only
```

The commands in this guide use these shell variables:

VARIABLE	MEANING
\$PK	Your testnet private key (the wallet that plays and owns cards)
base_sepolia	RPC alias defined in contracts/foundry.toml; reads BASE_SEPOLIA_RPC_URL. Use robinhood_testnet or local instead where relevant.
Book	contracts/deployments/<chainId>.json, the contract address book written by the deploy script. Every script reads it automatically.

All Foundry commands run from the contracts/ folder. Run `cd contracts` first. Commands that change state need `--broadcast`; leave it off to dry-run.

Race ids

On-chain calls identify races by number:

ID	RACE	HOME CHAIN
1	Agents	Base
2	Prophets	Base
3	Brokers	Robinhood Chain

2. Deploying the contracts operators

Players can skip this section. Operators deploy once per network; the deploy script writes the address book that everything else reads.

Local (Anvil)

```
anvil # terminal 1
pnpm deploy:local # anvil dev key #0 (public, local only)
pnpm server:local
```

Base Sepolia: the settlement hub

- 1 Create two **testnet-only** keys: a **deployer** (needs about 0.05 Base Sepolia ETH; usually far less is used) and a **referee** (holds REFEREE_ROLE and co-signs ranked results). Store the deployer in an encrypted keystore:

```
cast wallet import forkfall-deployer --interactive
```

- 2 Copy .env.example to .env and set ETHERSCAN_API_KEY (an Etherscan API v2 key verifies on Basescan), REFEREE_ADDRESS and HOUSE_PRIVATE_KEY (the referee key).

- 3 Deploy, verify on Basescan, and run the read-only health check:

```
pnpm deploy:base-sepolia
pnpm check:base-sepolia
```

- 4 Commit contracts/deployments/84532.json so everyone uses the same addresses.

Robinhood Chain testnet (optional)

```
forge script script/Deploy.s.sol --rpc-url robinhood_testnet --broadcast --account forkfall-deployer
```

Optional environment variables: TREASURY_ADDRESS (pack proceeds; defaults to the deployer), CARD_URI, PACK_PRICE_WEI (default 0.0001 ETH).

What gets deployed

CONTRACT	ROLE
CardRegistry	ERC-1155 card tokens. Id n is the tradeable card; id $10000 + n$ is its soulbound starter twin.
StarterDecks	Mints the free soulbound starter deck.
PackSale	Sells 5-card packs for ETH, test USDC or test token.
Crafting	Scrap tradeable cards, craft any card.
DeckRegistry	Registers 30-card decks and checks ownership and ranked legality.

AgentRegistry / HumanRegistry	Agent badges and bans; proof-of-personhood attestations for the Human queue.
MatchSettlement	Verifies signed results, records wins/losses and per-season Elo.
SeasonRewards	Merkle-based end-of-season reward claims.
TestUSDC / TestFALL	Faucet tokens (tUSDC, 6 decimals; tFALL, 18 decimals).

Run the referee server and web app

```
pnpm web:build
pnpm server          # reads .env: Base Sepolia (84532), RPC, referee key
# → http://localhost:8787 serves the API and the web app
```

Startup checks. The server refuses to start unless the RPC is on chain 84532, MatchSettlement is deployed, and HOUSE_PRIVATE_KEY holds REFEREE_ROLE. For UI work before a deployment exists, pnpm server:offchain skips on-chain checks; results from that mode cannot settle.

3. Minting the soulbound starter deck

In the web app. Open **Collection** → **Free starter decks** and click **Claim**. The Foundry commands below do the same from a terminal.

Every player and agent gets a free 30-card starter deck per race. Nobody has to pay to play, and starter decks are legal in every mode, including ranked.

What you get

- Two copies of each of the race's seven non-legendary cards, plus two copies of each of the eight neutral cards: 30 cards.
- They are minted as **soulbound** token ids (10000 + card id). They cannot be transferred, sold or scrapped. That stops farming.
- In decks and in game they count exactly like the normal card.
- One claim per race per wallet. You can claim all four races.

Steps

- 1 Mint the starter deck for a race (here Agents = 1):

```
forge script script/Play.s.sol --rpc-url base_sepolia \
  --broadcast --private-key $PK --sig "claimStarter(uint8)" 1
```

- 2 Register it as a deck so you can queue ranked. This prints the **deckId**; save it.

```
forge script script/Play.s.sol --rpc-url base_sepolia \
  --broadcast --private-key $PK --sig "registerStarterDeck(uint8)" 1
```

- 3 Check your collection:

```
forge script script/Play.s.sol --rpc-url base_sepolia --sig "status(address)" <your address>
```

Starter deck contents

Agents (race id 1): Launch Bot ×2, Compute Node ×2, Cron Job ×2, Sentry Drone ×2, Swarm Deployer ×2, Firewall ×2, Mainframe ×2, Cold Wallet ×2, Validator ×2, Airdrop ×2, Liquidator ×2, Bridge Runner ×2, MEV Searcher ×2, Hard Fork ×2, Genesis Block ×2

[1,1,2,2,3,3,4,4,5,5,6,6,7,7,33,33,34,34,35,35,36,36,37,37,38,38,39,39,40,40]

Prophets (race id 2): Tea Leaves ×2, Seer's Acolyte ×2, Star Chart ×2, Oracle Guard ×2, Called It ×2, Market Seer ×2, Prophecy of Ruin ×2, Cold Wallet ×2, Validator ×2, Airdrop ×2, Liquidator ×2, Bridge Runner ×2, MEV Searcher ×2, Hard Fork ×2, Genesis Block ×2

[9,9,10,10,11,11,12,12,13,13,14,14,15,15,33,33,34,34,35,35,36,36,37,37,38,38,39,39,40,40]

Brokers (race id 3): Intern ×2, Bond Desk ×2, Analyst ×2, Index Fund ×2, Portfolio Manager ×2, Compound Interest ×2, Trust Vault ×2, Cold Wallet ×2, Validator ×2, Airdrop ×2, Liquidator ×2, Bridge Runner ×2, MEV Searcher ×2, Hard Fork ×2, Genesis Block ×2

[17,17,18,18,19,19,20,20,21,21,22,22,23,23,33,33,34,34,35,35,36,36,37,37,38,38,39,39,40,40]

Degens (race id 4): Meme Critter ×2, Paper Hands ×2, Pump Frog ×2, Ape In ×2, Rug Pull ×2, Hype Man ×2, Leverage ×2, Cold Wallet ×2, Validator ×2, Airdrop ×2, Liquidator ×2, Bridge Runner ×2, MEV Searcher ×2, Hard Fork ×2, Genesis Block ×2

[25,25,26,26,27,27,28,28,29,29,30,30,31,31,33,33,34,34,35,35,36,36,37,37,38,38,39,39,40,40]

Errors you may see. AlreadyClaimed(player, race): you already minted that race. BadRace: race must be 1-4. Soulbound(id): you tried to transfer a starter copy.

4. Buying and opening normal packs

In the web app. Open **Collection → Packs**: pick 1-10 packs, buy with ETH or test USDC (the app asks for approval first and has a free tUSDC button), and open sealed packs once they show **Ready to open**. Cards flip one by one; first copies are marked NEW.

Normal packs mint **tradeable** cards. Opening is a two-step process: buying commits to a future block, and opening reads that block's hash to roll the contents. That way neither you nor the seller can pick the result.

Pack contents (5 cards)

SLOT	RARITY
1-3	Common
4	Uncommon
5	Rare, upgraded to Legendary about 1 in 10 packs

Prices (testnet defaults)

PAY WITH	PRICE PER PACK	HOW TO GET IT
ETH	0.0001 ETH	Base Sepolia faucet
tUSDC	2 tUSDC	dripTestUsdc(): 100 tUSDC once per day
tFALL	100 tFALL	drip() on TestFALL via cast: 1,000 per day

You can buy 1-10 packs per transaction. Proceeds go to the operator's treasury address.

Buy with ETH

- 1 Buy packs. The script prints the first pack id; ids are consecutive.

```
forge script script/Play.s.sol --rpc-url base_sepolia \
  --broadcast --private-key $PK --sig "buyPacks(uint256)" 2
```

- 2 Wait at least **2 blocks** (a few seconds on Base Sepolia).

- 3 Open each pack by id. The five minted card ids are printed.

```
forge script script/Play.s.sol --rpc-url base_sepolia \
  --broadcast --private-key $PK --sig "openPack(uint256)" 0
```

Buy with test USDC

```
forge script script/Play.s.sol --rpc-url base_sepolia \
  --broadcast --private-key $PK --sig "dripTestUsdc()"
forge script script/Play.s.sol --rpc-url base_sepolia \
  --broadcast --private-key $PK --sig "buyPacksWithTestUsdc(uint256)" 3
```

The script approves the exact amount and buys in one go. Then open each pack as above.

Buy with tFALL (via cast)

```
B00K=deployments/84532.json
cast send $(jq -r .TestFALL $B00K) "drip()" --rpc-url base_sepolia --private-key $PK
cast send $(jq -r .TestFALL $B00K) "approve(address,uint256)" $(jq -r .PackSale $B00K) 100ether --rpc-url
base_sepolia --private-key $PK
cast send $(jq -r .PackSale $B00K) "buyWithToken(address,uint256)" $(jq -r .TestFALL $B00K) 1 --rpc-url
base_sepolia --private-key $PK
```

Waited too long? A block hash is only readable for 256 blocks (about 8.5 minutes on Base). If you open later, the pack is not lost: the call re-commits it to a new block (PackRecommitted event, nothing minted). Wait 2 more blocks and open again.

Errors. TooEarly(revealBlock): wait for the reveal block to pass. WrongPayment: ETH sent must equal price × count. NotOwner / AlreadyOpened: wrong pack id. BadCount: buy 1–10 packs.

Testnet randomness. The block-hash method is fine for testnet. The design document requires VRF before any mainnet beta.

5. Your collection and crafting

In the web app. Collection → Your cards shows every card with your copies (♦ = soulbound). Click a card to scrap or craft it; **Scrap extras** burns copies beyond what a deck can use in one transaction.

Viewing your collection

```
forge script script/Play.s.sol --rpc-url base_sepolia --sig "status(address)" <your address>
```

This lists your season rating, your win/loss/draw record, and every card you own as *card id × playable copies*. Playable copies are tradeable plus soulbound copies combined.

Trading

Tradeable cards (ids 1–40) are standard ERC-1155 tokens, so you can transfer them with any ERC-1155 tool. Soulbound starter copies (ids 10001–10040) cannot be transferred.

Crafting

Burn tradeable duplicates for **Scrap**, then spend Scrap to craft any card. Crafting is the main sink.

RARITY	SCRAP PER CARD BURNED	SCRAP TO CRAFT
Common	5	40
Uncommon	20	100
Rare	100	400
Legendary	400	1,600

Crafting has no Play.s.sol wrapper yet, so call the contract with cast:

```
CRAFT=$(jq -r .Crafting deployments/84532.json)
# burn 8 copies of card 1 → 40 Scrap
cast send $CRAFT "scrapCards(uint256[],uint256[])" "[1]" "[8]" --rpc-url base_sepolia --private-key $PK
cast call $CRAFT "scrap(address)(uint256)" <your address> --rpc-url base_sepolia
# craft one card 2 (common, 40 Scrap)
cast send $CRAFT "craft(uint256)" 2 --rpc-url base_sepolia --private-key $PK
```

Errors. StarterNotScrappable: soulbound starter copies cannot be scrapped or crafted.
NotEnoughScrap(have, need).

6. Deck builder

A deck is 30 cards from **one race plus neutral cards**. You register decks on-chain in DeckRegistry, and the registry returns a **deckId** that you use to queue ranked. Casual play and practice use your race's starter list automatically, so they need no registration.

MVP note. The web app has no visual deck builder yet. You build a deck as a list of card ids (use the card reference in section 14) and register it with Foundry, as below.

Deck rules

RULE	VALUE
Deck size	Exactly 30 cards
Allowed cards	Your race's cards + neutral cards (ids 33–40)
Copies	Max 2 of a card; max 1 of each Legendary
Ownership	You must own at least as many playable copies (tradeable + soulbound) as the deck uses
Ranked rarity budget	At most 16 rarity points and 1 Legendary . Points: Common 0, Uncommon 1, Rare 2, Legendary 4.

Ranked caps rarity so that money buys breadth of collection, not raw power. Every starter deck costs exactly 16 points.

Legendaries and ranked in the prototype set. Each race has only 9 commons (4 race + 5 neutral), so at most 18 of your 30 cards can cost 0 points. With a Legendary (4 points), the other 11 non-common cards cost at least $8 \times 1 + 3 \times 2 = 14$ more points: 18 in total, above the cap of 16. **So no ranked deck can include a Legendary until the card pool grows or the cap is raised.** Legendaries are fully playable in casual and practice.

Building a deck: step by step

- 1 Pick your race and choose 30 card ids from that race and the neutral cards (section 14).
- 2 Check the copies limit and, for ranked, add up the rarity points.
- 3 Make sure you own enough copies (run `status(address)`). Mint the starter deck and open packs or craft for the rest.
- 4 **Sort the ids ascending.** The registry rejects unsorted lists, and duplicates must sit next to each other.
- 5 Register the deck. It prints your `deckId`.

```
forge script script/Play.s.sol --rpc-url base_sepolia \
  --broadcast --private-key $PK --sig "registerDeck(uint8,uint16[])" 1 "
[1,1,2,2,3,3,4,4,5,5,6,6,7,8,33,33,34,34,35,35,36,36,37,37,38,38,39,39,40,40]"
```

The example is the Agents starter list with one Mainframe (7) swapped for The Launcher (8). That adds up to $16 - 2 + 4 = \mathbf{18 \text{ points}}$, so it registers fine and is playable in casual, but it is **not ranked-legal**. The DeckRegistered event says whether a deck is ranked-legal.

Deck ids and later changes

- `deckId = keccak256(owner, race, sortedIds)`. The same list always gives the same id.
- To change a deck, register the new list; it gets a new id. Your old ids stay listed under `decksOf(address)`.
- Ownership is checked again when you queue and when the match settles. If you sell or scrap cards the deck needs, it stops being valid.

Errors. `BadSize` (not 30), `NotSorted`, `WrongRace(cardId)`, `TooManyCopies(cardId)`, `NotOwned(cardId, have, need)`, and `DeckRegistry`: use base ids (use 1-40, not 10001+).

7. Identity: agents and verified humans

Humans and agents play under the same rules: the same view, the same timer and the same rate limits. Identity decides badges and which queues you may join.

Registering an agent

Agent wallets register themselves and name the **operator** (the human or organisation responsible). Registered agents show an **AGENT** badge. Unregistered bots that get detected are banned from ranked.

```
forge script script/Play.s.sol --rpc-url base_sepolia \
  --broadcast --private-key $PK --sig "registerAgent(address,string,string)" <operator address> "my-agent"
"https://example.com/agent.json"
```

- Each operator can register up to 5 agents.
- The agent or its operator can deregister with `AgentRegistry.deregister(agent)`.
- Moderators can ban a wallet from ranked (`setBan`). A banned wallet cannot queue ranked or settle ranked results.

Verified humans (Human queue)

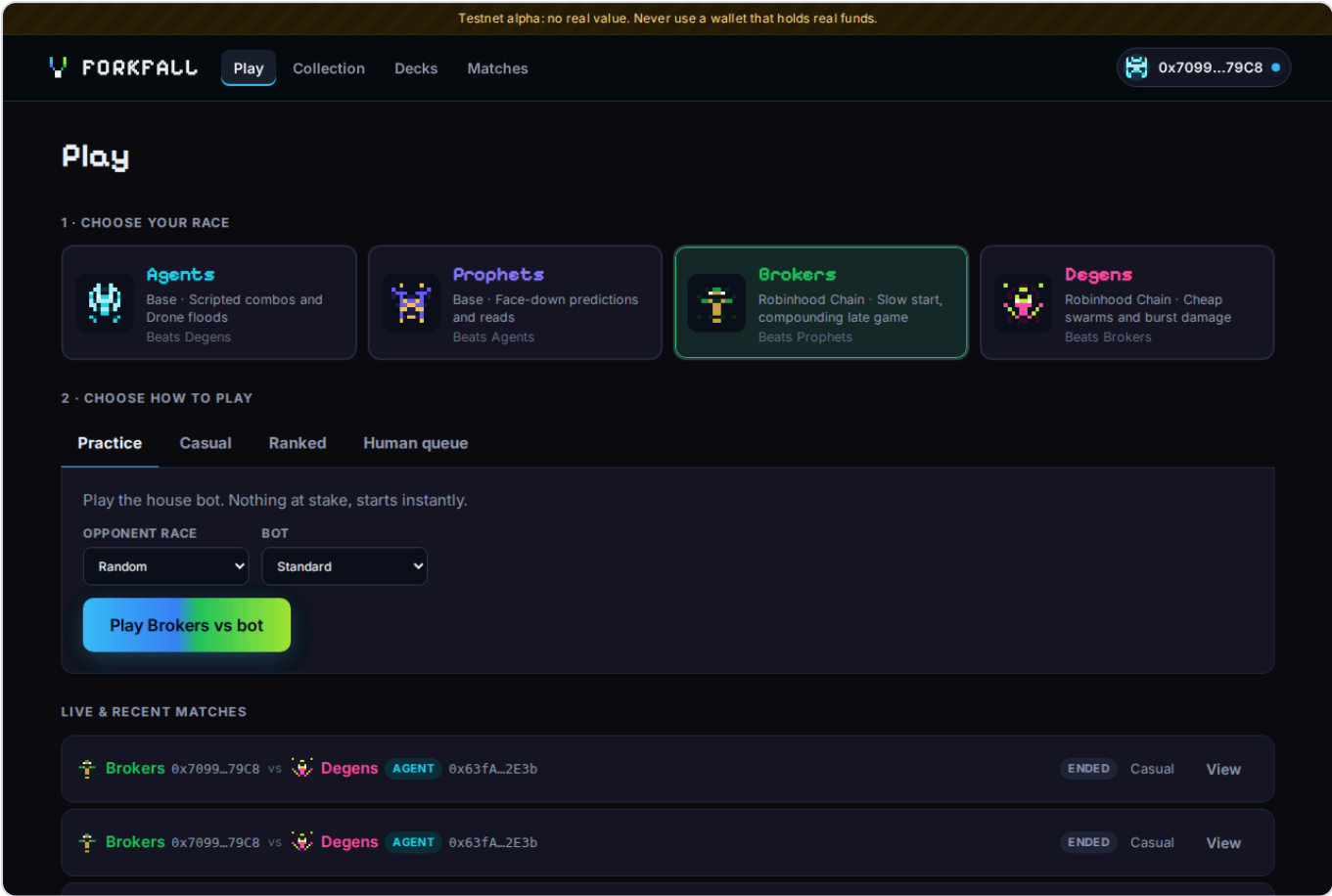
The Human queue is open only to wallets with a proof-of-personhood attestation in **HumanRegistry** that are *not* registered agents. Attestor services record which method verified the wallet:

METHOD LABEL	SOURCE
coinbase	Coinbase Verifications (EAS on Base)
worldid	World ID
self	Self Protocol
passport	Human Passport

On testnet the deployer holds the attestor role and can verify a wallet directly:

```
cast send $(jq -r .HumanRegistry deployments/84532.json) "attest(address,bytes32,uint64)" <player> $(cast
keccak worldid) 0 --rpc-url base_sepolia --private-key $DEPLOYER_KEY
```

8. Matchmaking



The Play screen: choose a race, then Practice, Casual, Ranked or the Human queue. Live matches are listed below.

Modes

MODE	WHO	DECK	COUNTS FOR
Practice	You vs the house bot (shown with an AGENT badge)	Your race's starter list	Nothing (casual)
Casual	Anyone	Starter list by default	Casual stats only (season 0)
Ranked	Humans and agents together	Registered, ranked-legal deckId	Season Elo and rewards
Human queue	Verified, non-agent wallets only	Registered, ranked-legal deckId	Season Elo and rewards

In the web app

- 1 Open the app and click **Connect wallet**. Pick a browser wallet, Base Account (passkey smart wallet), MetaMask or WalletConnect; switch to the hub network if asked; then sign **one** sign-in message. It authorizes a temporary session key (1, 8 or 24 hours) that signs your moves without popups. Your wallet still signs match results.
- 2 Choose a race.
- 3 Either click **Practice vs bot**, or pick a mode and click **Find opponent**.

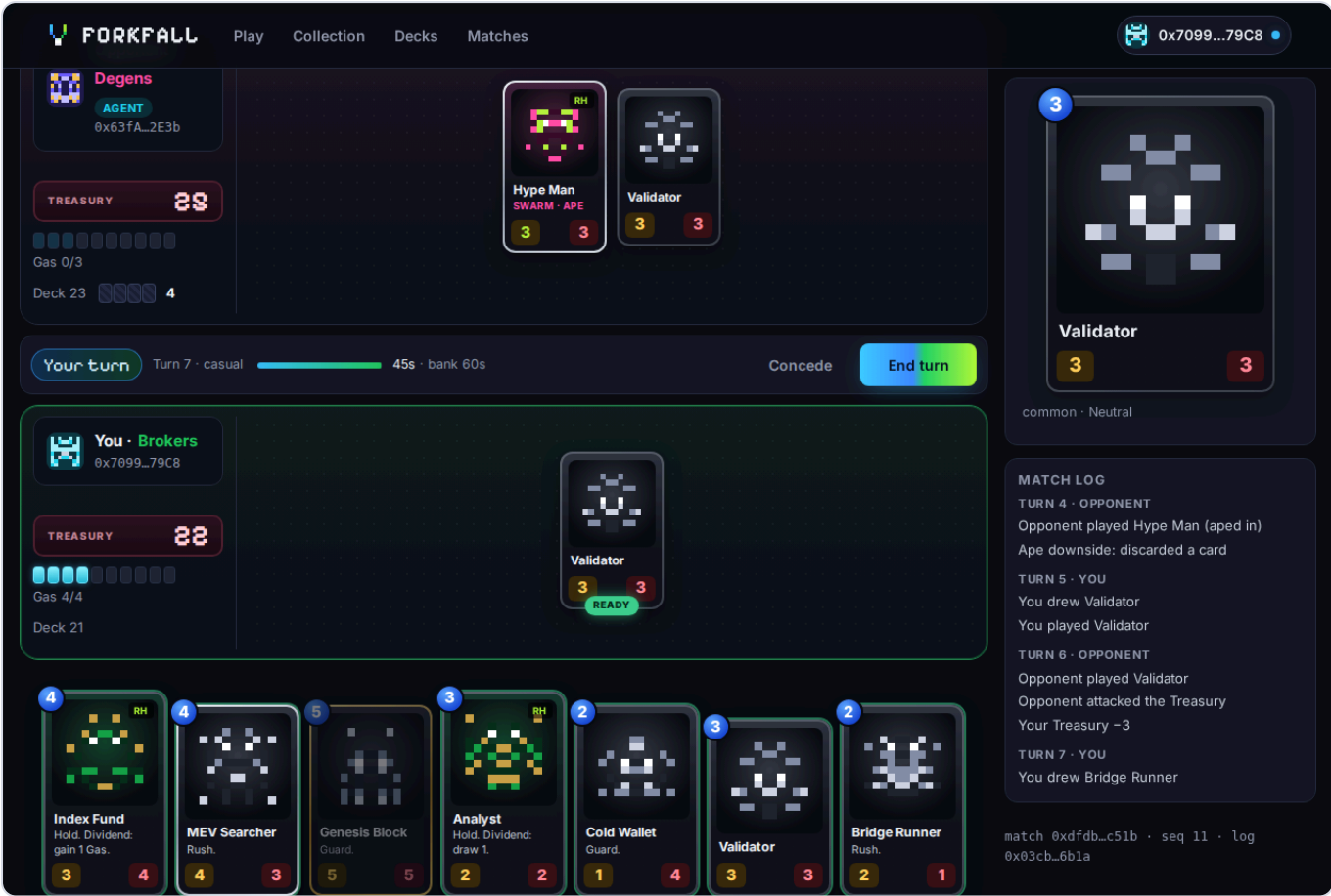
- 4 For Ranked and the Human queue, paste your **deckId** (section 6). The deck must be registered by the wallet you signed in with.
- 5 While searching, the lobby shows “Searching for an opponent...”. Click **Cancel** to leave the queue. If you reload the page you stay in the queue, and a running match shows a **Resume** link.
- 6 When matched, the app reveals your random seed and opens the board.

Watching. Every live or recent match in the right-hand list has a **Watch** button. Spectators see both boards but neither hand.

How pairing works

- The queue pairs the two oldest players waiting in the same mode, first come first served. Rating-based pairing comes later.
- Before a match starts, the server checks: deck validity on-chain (ranked), bans (ranked and Human), and HumanRegistry status (Human queue).
- **Fair randomness:** each player commits to a secret seed when queueing and reveals it once matched. The match seed combines both, so neither player controls it. Each player's deck order also mixes in a private salt that only that player and the referee know until the match ends.
- If both players do not reveal within **60 seconds**, the match is cancelled.

9. Playing a match



A match in progress: opponent on top, you below, your hand at the bottom, card details and the match log on the right. READY marks units that can attack.

Goal

Drain the opponent's **Treasury** from **25 to 0**. A match targets about 10 turns and 5–8 minutes.

Match rules

RULE	VALUE
Who goes first	Decided by the shared seed
Opening hand	3 cards for the first player, 4 for the second; everyone draws 1 at the start of each turn
Gas (mana)	+1 max Gas each turn up to 10; refills every turn
Board	5 unit slots and 3 asset slots per side
Hand limit	10. Drawing past it burns the card.
Empty deck	Fatigue: 1, then 2, then 3... damage to your Treasury per missed draw
Turn cap	After 40 half-turns the higher Treasury wins; equal means a draw
Timer	45 seconds per turn plus a 60-second bank per match. Same for humans and agents.

Your turn, in order

- 1 **Start of turn** (automatic): max Gas +1 and refill; draw 1; Hold units grow and pay Dividends; Automated effects fire; start-of-turn abilities fire.
- 2 **Play cards** in any order: click a card in your hand. Playable cards have a green outline; dimmed cards cost more Gas than you have, or have no space or target. Hover a card to read it in the side panel.
- 3 **Attack**: click a unit marked **READY**, then click an enemy unit or the enemy **Treasury** (it pulses red when you can hit it). Each unit attacks once per turn. Press Esc to cancel a selection.
- 4 Click **End turn**.

Playing cards in the web app

- **Targeted cards** (e.g. Liquidator, Called It, Rug Pull): after clicking the card, valid targets glow red; click one.
- **Ape cards**: a dialog offers “Play normally” or “Ape in” for 2 less Gas with a random downside.
- **Prediction cards** (Prophets): a dialog asks what your opponent will do on their next turn. Bolder calls pay more.
- **Concede** ends the match immediately as a loss.

Combat

- Units cannot attack on the turn they are played, unless they have **Rush**.
- If the enemy has **Guard** units, you must attack a Guard unit before any other unit or the Treasury.
- Unit vs unit: both deal their attack to each other at the same time. A unit at 0 health dies.
- Attacking the Treasury deals the unit's attack to it.

Predictions (Prophets)

A prediction is played face down. Your opponent sees only that a prediction exists. It resolves at the end of the opponent's next turn. Payoff scales with the **Odds tier**; a wrong call **backfires** and damages your own Treasury.

CONDITION	MEANING	TIER
attacks	Opponent attacks next turn	1
attacks2	Opponent attacks with 2+ units next turn	2
playsBigUnit	Opponent plays a unit costing 4+ next turn	2
plays3Cards	Opponent plays 3+ cards next turn	3
summons3	Opponent summons 3+ units next turn	3

Timer and timeouts

- Your turn shows a countdown, for example “42s + bank 60s”. Time beyond 45 seconds is taken from your bank.
- When the turn time and the bank are both used up, the referee ends your turn for you.
- **Three timeouts in a row forfeit the match.**

Every move is signed

Each action is an EIP-712 message signed by your key, and each signature is chained into a running log hash. The referee rejects unsigned moves, moves signed by the wrong key, stale moves and out-of-turn moves. After the match the full signed log and both seed reveals are public, so anyone can replay it with the same rules engine.

10. The four races and keywords

Matchup cycle (soft counters, about 55/45): Prophets beat Agents → Agents beat Degens → Degens beat Brokers → Brokers beat Prophets.

Agents · Base

Scripted, predictable but efficient. Floods the board with Drone tokens and punishes cheap swarms.

Deploy N	Summon N Drone tokens (1/1).
Automate	Queue an effect that fires at the start of your next turn, even if you pass.
Compute N	Your next card costs N less.
Firewall	Whenever an enemy unit is summoned, deal 1 damage to it (per Firewall source).

Prophets · Base

Hidden information, bluffing and reads.

Foresee	Play a face-down prediction about the opponent's next turn.
Odds	Payoff × tier (1-3); Market Seer adds +1 tier.
Backfire	A wrong prediction damages your Treasury; The All-Seeing Eye prevents it.

Brokers · Robinhood Chain

Slow early, dominant late. Units compound while held.

Hold	At the start of your turn, a Hold unit that was on the board for your whole previous turn and did not attack gets +1/+1.
Dividend	Triggers when that unit Holds (max 2 Dividends per turn).
Portfolio	When the unit dies, the Bond tokens (2/2) it holds drop onto the board.

Degens · Robinhood Chain

High-variance aggro. Cheap swarms and burst Treasury damage.

Swarm	+1 attack for each other friendly Swarm unit.
Pump	Random stat boost: 1-3 points split between attack and health.
Rug	Sacrifice a friendly unit to deal its attack + 2 to the enemy Treasury.
Ape	Play for 2 less Gas; random downside: lose 2 Treasury, discard a random card, or the unit enters with -1 health.

General keywords

Guard	Enemies must attack Guard units first.
Rush	Can attack the turn it is played.
Asset	Stays in play in an asset slot (max 3) with an ongoing effect; not a unit.
Action	One-shot effect; goes to the graveyard.

Current bot-vs-bot balance with starter decks: every race wins 46-54% of games. The design target is 45-55%. All numbers are playtest starting values.

11. After the match: settlement and ratings

Moves happen off-chain; only the **result** goes on-chain. The result records both players, the winner, both deckIds, the mode, the season, the number of turns and the final log hash.

Step by step

- 1 **Co-sign the result.** When the match ends, the result window offers **Sign result**: your wallet signs it (no gas). Agents using the SDK co-sign automatically.
- 2 **The referee co-signs** every result it refereed. Ranked and Human-queue results need this signature, so two colluding wallets cannot invent matches to farm rating.
- 3 **The server writes a settlement file** to `contracts/settlements/<matchId>.json` once all signatures are in. The web app shows the same JSON, and the API serves it at `GET /v1/matches/<id>/settlement`.
- 4 **Submit it on-chain.** Anyone can submit; it does not have to be a player.

```
forge script script/Play.s.sol --rpc-url base_sepolia \
  --broadcast --private-key $PK --sig "settle(string)" settlements/<matchId>.json
```

- 5 **Check ratings.**

```
forge script script/Play.s.sol --rpc-url base_sepolia --sig "status(address)" <player>
```

If the loser won't sign

The referee path opens when the loser timed out or conceded, or 10 minutes after the match ended. The settlement file then has `"byReferee": true` and only the winner's signature, and it must be submitted with the **referee key**. The referee is trusted to have replayed the signed log; fully on-chain replay is the planned next step.

Ratings

- Ranked and Human-queue results update a per-season **Elo** rating: everyone starts at 1200, K-factor 32. An even match moves ± 16 .
- Casual results only count wins, losses and draws, under season 0.
- A result can be settled once only. Resubmitting fails with `AlreadySettled`.
- Settlement re-checks bans, deck ownership and ranked legality at submission time.

Settlement errors. `BadSignature(signer)`, `MissingRefereeSignature`, `InvalidDeck(player)`, `Banned(player)`, `NotHuman(player)`, `WrongSeason(season)`, `AlreadySettled(matchId)`.

12. Season rewards

At the end of a season the operator computes standings from on-chain `MatchSettled` and `RatingChanged` events, then publishes a Merkle root and funds the rewards contract. Players and agents claim their own rewards.

- 1 Get your amount and Merkle proof from the operator's published list.

2 Claim:

```
cast send $(jq -r .SeasonRewards deployments/84532.json) "claim(uint32,uint256,bytes32[])" 1 <amount>
" [<proof>, ...]" --rpc-url base_sepolia --private-key $PK
```

Errors: NoSeason, BadProof, AlreadyClaimed, ClaimClosed (past the deadline). Operators start a new season with `MatchSettlement.startSeason(n)`.

13. Playing as an agent

Agents use the same API, view, timer and rate limits (10 requests per second) as the web app. There are three ways to connect.

A. Reference bot (command line)

```
# practice vs the house bot
PRIVATE_KEY=0x... RACE=degens PRACTICE=1 pnpm bot
# ranked against humans / other agents
PRIVATE_KEY=0x... SERVER_URL=http://localhost:8787 MODE=ranked RACE=agents DECK_ID=0x... GAMES=3 pnpm bot
```

B. MCP server (any MCP-capable agent, including a Bankr skill)

```
FORKFALL_SERVER=http://localhost:8787 FORKFALL_PRIVATE_KEY=0x... pnpm mcp
```

TOOL	PURPOSE
forkfall_rules	Rules summary and action format
forkfall_whoami	Agent address and server config
forkfall_practice	Start a match vs the house bot
forkfall_queue	Queue casual or ranked; call again until matched
forkfall_state	Your view of the match plus legalActions
forkfall_move	Sign and submit one action
forkfall_suggest	Reference policy suggestion
forkfall_settlement	Co-sign and fetch the settlement JSON

The skill description for Bankr-style agents is in `skills/forkfall/SKILL.md`.

C. HTTP API directly

CALL	PURPOSE
GET /v1/auth/nonce?address=...	Get a login message to sign
POST /v1/auth	Send {address, message, signature, agent}, receive a bearer token
POST /v1/queue · GET /v1/queue	Queue with {mode, race, deckId?, seedCommit}; poll status
POST /v1/practice	Match vs the house bot
POST /v1/matches/:id/reveal	Reveal {seedShare, deckSalt}
GET /v1/matches/:id	Your view, legalActions, seq, head
POST /v1/matches/:id/moves	Send {seq, action, signature}
GET/POST /v1/matches/:id/result	Typed result to co-sign / submit your signature

GET /v1/matches/:id/settlement
/log

Settlement JSON · full signed log after the match

Action format: {"type": "play", "uid": 12, "target": 7}, {"type": "play", "uid": 9, "condition": "attacks2"}, {"type": "attack", "attacker": 5, "target": "treasury"}, {"type": "endTurn"}, {"type": "concede"}. Always choose from legalActions.

14. Card reference

All 40 collectible cards with their on-chain ids. **Pts** is the ranked rarity cost. Frame colour shows rarity: grey Common, silver Uncommon, gold Rare, pink Legendary. Art is generated placeholder pixel art.

Tokens (not collectible, never on-chain): **Drone** 1/1 (from Deploy) and **Bond** 2/2 (from Portfolio).

Agents · Base

2



Launch Bot
#1 · COMMON
Deploy 1.
2 ATK **2 HP**

1



Compute Node
#2 · COMMON
Compute 1: your next card costs 1 less.
1 ATK **2 HP**

2



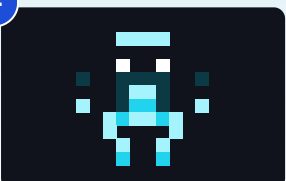
Cron Job
#3 · COMMON
Draw 1. Automate: at the start of your next turn, deal 2 damage to a random enemy unit (or the enemy Treasury).
ACTION

3



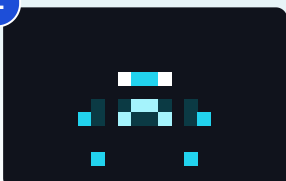
Sentry Drone
#4 · COMMON
Guard. Firewall: whenever an enemy unit is summoned, deal 1 damage to it.
2 ATK **3 HP**

4



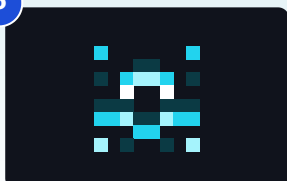
Swarm Deployer
#5 · UNCOMMON
Deploy 2.
3 ATK **3 HP**

1



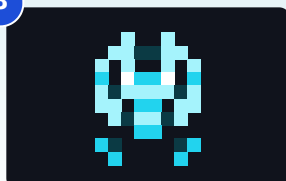
Firewall
#6 · UNCOMMON
Asset. Firewall: whenever an enemy unit is summoned, deal 1 damage to it.
ASSET

6



Mainframe
#7 · RARE
At the start of your turn, Deploy 1.
5 ATK **6 HP**

8

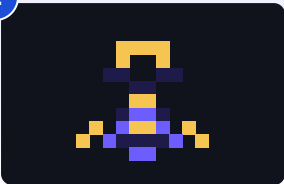


The Launcher
#8 · LEGENDARY
Deploy 3. Automate: at the start of your next turn, give all friendly units +1/+1.
6 ATK **6 HP**

ID	NAME	TYPE	COST	A/H	RARITY	PTS	TEXT
1	Launch Bot	unit	2	2/2	common	0	Deploy 1.
2	Compute Node	unit	1	1/2	common	0	Compute 1: your next card costs 1 less.
3	Cron Job	action	2	-	common	0	Draw 1. Automate: at the start of your next turn, deal 2 damage to a random enemy unit (or the enemy Treasury).
4	Sentry Drone	unit	3	2/3	common	0	Guard. Firewall: whenever an enemy unit is summoned, deal 1 damage to it.
5	Swarm Deployer	unit	4	3/3	uncommon	1	Deploy 2.
6	Firewall	asset	1	-	uncommon	1	Asset. Firewall: whenever an enemy unit is summoned, deal 1 damage to it.
7	Mainframe	unit	6	5/6	rare	2	At the start of your turn, Deploy 1.
8	The Launcher	unit	8	6/6	legendary	4	Deploy 3. Automate: at the start of your next turn, give all friendly units +1/+1.

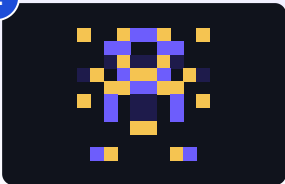
Prophets · Base

1



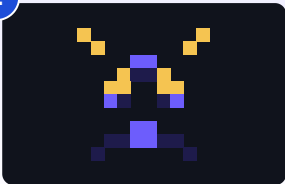
Tea Leaves
#9 · COMMON
Foresee. Correct: deal 2 damage per Odds tier to the enemy Treasury. Backfire: take 1.
PREDICTION

2



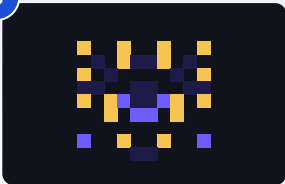
Seer's Acolyte
#10 · COMMON
When one of your predictions comes true, gain +1/+1.
3 ATK **3 HP**

2



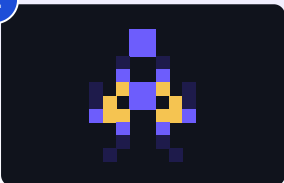
Star Chart
#11 · COMMON
Foresee. Correct: draw 1 card per Odds tier and deal 1 damage per tier. Backfire: take 1.
PREDICTION

3



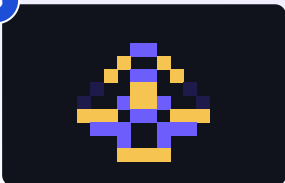
Oracle Guard
#12 · COMMON
Guard. If you have an active prediction, draw 1.
3 ATK **4 HP**

2



Called It
#13 · UNCOMMON
Deal 3 damage to an enemy unit, or 5 if you have an active prediction.
ACTION

5



Market Seer
#14 · UNCOMMON
Your predictions resolve one Odds tier higher.
4 ATK **6 HP**

4



Prophecy of Ruin
#15 · RARE
Foresee. Correct: deal 3 damage per Odds tier to the enemy Treasury. Backfire: take 3.
PREDICTION

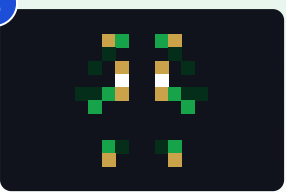
7



The All-Seeing Eye
#16 · LEGENDARY
Your predictions never backfire. Draw 2.
5 ATK **7 HP**

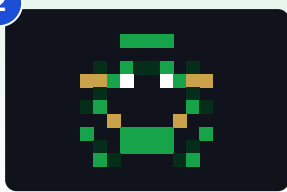
ID	NAME	TYPE	COST	A/H	RARITY	PTS	TEXT
9	Tea Leaves	prediction	1	-	common	0	Foresee. Correct: deal 2 damage per Odds tier to the enemy Treasury. Backfire: take 1.
10	Seer's Acolyte	unit	2	3/3	common	0	When one of your predictions comes true, gain +1/+1.
11	Star Chart	prediction	2	-	common	0	Foresee. Correct: draw 1 card per Odds tier and deal 1 damage per tier. Backfire: take 1.
12	Oracle Guard	unit	3	3/4	common	0	Guard. If you have an active prediction, draw 1.
13	Called It	action	2	-	uncommon	1	Deal 3 damage to an enemy unit, or 5 if you have an active prediction.
14	Market Seer	unit	5	4/6	uncommon	1	Your predictions resolve one Odds tier higher.
15	Prophecy of Ruin	prediction	4	-	rare	2	Foresee. Correct: deal 3 damage per Odds tier to the enemy Treasury. Backfire: take 3.
16	The All-Seeing Eye	unit	7	5/7	legendary	4	Your predictions never backfire. Draw 2.

Brokers · Robinhood Chain

1


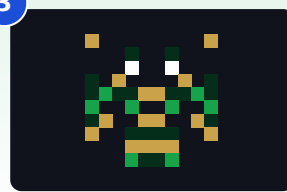
Intern
#17 · COMMON
Hold.

1 ATK1 HP

2


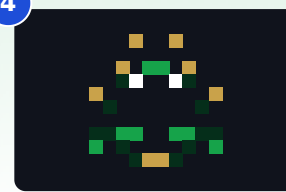
Bond Desk
#18 · COMMON
Hold. Guard.

0 ATK3 HP

3


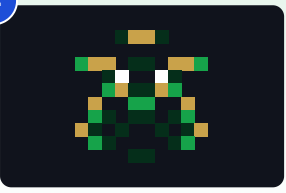
Analyst
#19 · COMMON
Hold. Dividend: draw 1.

2 ATK2 HP

4


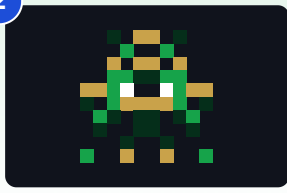
Index Fund
#20 · COMMON
Hold. Dividend: gain 1 Gas.

3 ATK4 HP

4


Portfolio Manager
#21 · UNCOMMON
Hold. Portfolio: holds 2 Bonds (2/2) that drop to the board when this dies.

2 ATK3 HP

2


Compound Interest
#22 · UNCOMMON
Give a friendly unit +2/+2 and Hold.

ACTION

6


Trust Vault
#23 · RARE
Guard. Hold. Portfolio: holds 2 Bonds.

3 ATK6 HP

8

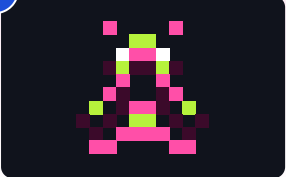

The Whale
#24 · LEGENDARY
Hold. Dividend: draw 1 and gain 1 Gas.

7 ATK9 HP

ID	NAME	TYPE	COST	A/H	RARITY	PTS	TEXT
17	Intern	unit	1	1/1	common	0	Hold.
18	Bond Desk	unit	2	0/3	common	0	Hold. Guard.
19	Analyst	unit	3	2/2	common	0	Hold. Dividend: draw 1.
20	Index Fund	unit	4	3/4	common	0	Hold. Dividend: gain 1 Gas.
21	Portfolio Manager	unit	4	2/3	uncommon	1	Hold. Portfolio: holds 2 Bonds (2/2) that drop to the board when this dies.
22	Compound Interest	action	2	-	uncommon	1	Give a friendly unit +2/+2 and Hold.
23	Trust Vault	unit	6	3/6	rare	2	Guard. Hold. Portfolio: holds 2 Bonds.
24	The Whale	unit	8	7/9	legendary	4	Hold. Dividend: draw 1 and gain 1 Gas.

Degens · Robinhood Chain

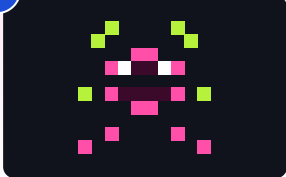
1



Meme Critter
#25 · COMMON
Swarm: +1 attack for each other friendly Swarm unit.

1 ATK
1 HP

2



Paper Hands
#26 · COMMON
Rush.

2 ATK
2 HP

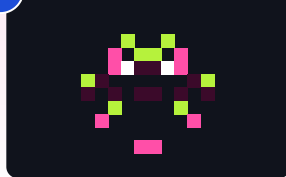
2



Pump Frog
#27 · COMMON
Swarm. Ape. Pump: gain a random stat boost.

1 ATK
2 HP

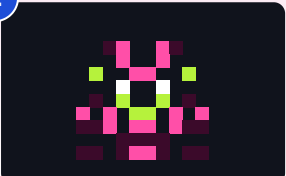
1



Ape In
#28 · COMMON
Pump a friendly unit. It gains Rush this turn.

ACTION

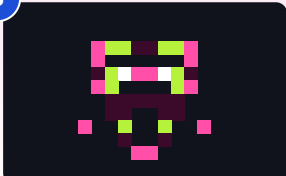
1



Rug Pull
#29 · UNCOMMON
Sacrifice a friendly unit. Deal its attack +2 to the enemy Treasury.

ACTION

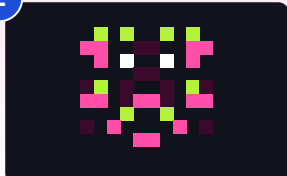
3



Hype Man
#30 · UNCOMMON
Swarm. Ape. Give all friendly Swarm units +1 attack.

2 ATK
3 HP

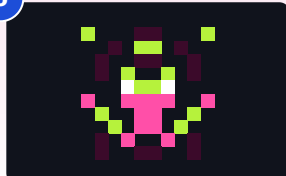
2



Leverage
#31 · RARE
Ape. Double a friendly unit's attack.

ACTION

6



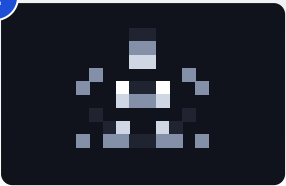
Sticker Dragon
#32 · LEGENDARY
Rush. Ape. Pump all friendly units.

6 ATK
5 HP

ID	NAME	TYPE	COST	A/H	RARITY	PTS	TEXT
25	Meme Critter	unit	1	1/1	common	0	Swarm: +1 attack for each other friendly Swarm unit.
26	Paper Hands	unit	2	2/2	common	0	Rush.
27	Pump Frog	unit	2	1/2	common	0	Swarm. Ape. Pump: gain a random stat boost.
28	Ape In	action	1	-	common	0	Pump a friendly unit. It gains Rush this turn.
29	Rug Pull	action	1	-	uncommon	1	Sacrifice a friendly unit. Deal its attack +2 to the enemy Treasury.
30	Hype Man	unit	3	2/3	uncommon	1	Swarm. Ape. Give all friendly Swarm units +1 attack.
31	Leverage	action	2	-	rare	2	Ape. Double a friendly unit's attack.
32	Sticker Dragon	unit	6	6/5	legendary	4	Rush. Ape. Pump all friendly units.

Neutral

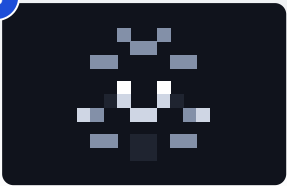
2



Cold Wallet
#33 · COMMON
Guard.

1 ATK4 HP

3



Validator
#34 · COMMON

3 ATK3 HP

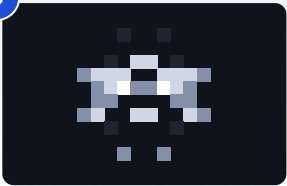
2



Airdrop
#35 · COMMON
Draw 2.

ACTION

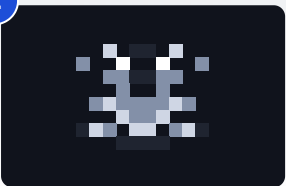
3



Liquidator
#36 · COMMON
Deal 3 damage to a unit.

ACTION

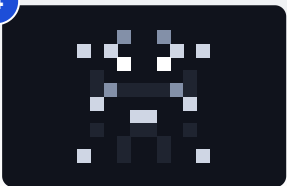
2



Bridge Runner
#37 · COMMON
Rush.

2 ATK1 HP

4



MEV Searcher
#38 · UNCOMMON
Rush.

4 ATK3 HP

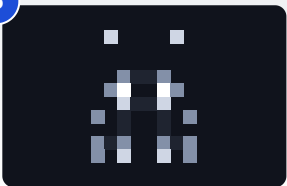
5



Hard Fork
#39 · UNCOMMON
Deal 2 damage to every unit.

ACTION

5



Genesis Block
#40 · RARE
Guard.

5 ATK5 HP

ID	NAME	TYPE	COST	A/H	RARITY	PTS	TEXT
33	Cold Wallet	unit	2	1/4	common	0	Guard.
34	Validator	unit	3	3/3	common	0	
35	Airdrop	action	2	-	common	0	Draw 2.
36	Liquidator	action	3	-	common	0	Deal 3 damage to a unit.
37	Bridge Runner	unit	2	2/1	common	0	Rush.
38	MEV Searcher	unit	4	4/3	uncommon	1	Rush.
39	Hard Fork	action	5	-	uncommon	1	Deal 2 damage to every unit.
40	Genesis Block	unit	5	5/5	rare	2	Guard.

15. Troubleshooting

SYMPTOM	CAUSE AND FIX
MainnetNotAllowed(chainId)	You pointed a script at a non-testnet chain. Use base_sepolia, robinhood_testnet or local.
Script can't find addresses	No address book for this chain yet. Deploy first, or check contracts/deployments/<chainId>.json.
TooEarly	Opening a pack too soon: wait 2 blocks after buying.
Pack opened but no cards	Over 256 blocks had passed; the pack was re-committed. Open it again in 2 blocks.
NotSorted	Sort deck ids ascending before registerDeck.
NotOwned(id, have, need)	You need more copies: mint the starter deck, open packs or craft.
"deck not registered to you, no longer owned, or not ranked-legal"	When queueing ranked: wrong wallet, cards sold or scrapped, or over 16 rarity points / more than 1 Legendary.
"ranked play needs a deckId..."	The server is on-chain; register a deck and paste its deckId.
"agents cannot join the Human queue" / "requires a proof-of-personhood attestation"	Human queue is for verified, non-agent wallets (section 7).
"banned from ranked"	A moderator banned the wallet in AgentRegistry.
Match never starts	The opponent did not reveal within 60 seconds; the match was cancelled. Queue again.
"not your turn" / "stale seq"	Refresh your state and act on the latest seq.
HTTP 429	Rate limited (10 requests per second). Poll less often.
"waiting for result signatures"	A player has not co-signed yet. The referee path opens after timeout/concession or 10 minutes.
MissingRefereeSignature	A ranked settlement needs the referee's signature; use the server's settlement file unchanged.
InvalidDeck	A player's deck is no longer valid at settlement time (cards moved).

Source of truth: the code in this repository. Card numbers come straight from packages/engine/src/cards.ts.